SPARK HOMES

# Developer Contest

## Project Brief

Prepared by Spark Homes • June 2026

| PRIZE | APPLICANTS | DEADLINE |
|---|---|---|
| **$1,000 Cash** | **1,600+** | **7/14/2026** |
| Top 10 earn interviews | All invited to compete | Late submissions not reviewed |

## Who We Are

Spark Homes is a residential real estate investment company based in the Tulsa County area. We buy distressed, outdated, or undervalued single family homes, renovate them, and resell them at a profit. This is commonly known as house flipping, and it is our core business.

A typical deal moves through three phases:

| | |
|---|---|
| **1. Acquisition** | We identify a distressed property and evaluate whether the numbers work. This means estimating what the home will sell for after repairs (the ARV — After Repair Value) and subtracting what repairs will cost. If the margin is there, we make an offer. |
| **2. Renovation** | Once we own the property, our contractors go to work. Repairs range from cosmetic (paint, flooring, fixtures) to structural (foundation, roof, full electrical rewires). The scope and cost of that work was locked in at acquisition. |
| **3. Resale** | We list the renovated home, sell it, and collect the margin between our total cost basis and the sale price. Speed matters  every extra week we hold the property costs money in financing, taxes, and utilities. |

## The Problem This App Solves

The most important number in our business is the repair estimate. Get it right and the deal makes money. Get it wrong and we absorb the difference out of our profit margin.

Our acquisition agents walk through properties before we make an offer. They are standing in an empty house, often without great cell service, and they need to answer one question as fast and accurately as possible:

> ***"How much will it cost to fix this place up?"***

Today, agents rely on memory, rough notes, and spreadsheets that are not built for mobile use. The process is slow, inconsistent between agents, and easy to miscalculate. A single missed line item, a furnace that needs replacing, or a bathroom that needs a full tile tearout can swing the estimate by thousands of dollars. The app we are asking you to build replaces that process. It gives every agent a fast, structured checklist they can run through room by room on their phone. Every line item has a standard unit cost already loaded. The agent selects what needs work, enters a quantity, and the app calculates a running total in real time. When the walkthrough is done, they export the estimate to Excel and share it with the team.

We built a working version internally. Now we want to see if someone can build it better.

## Executive Summary

We received over 1,600 applications for a developer role at Spark Homes and cannot fairly evaluate every resume at that volume. Rather than filter on credentials, we are inviting every applicant to compete on the same take-home project that mirrors the actual job.

The project: build a mobile-first repair cost estimator web application for use by our acquisition team in the field.

The candidate who ships the best version wins $1,000 cash. The top 10 submissions each earn a final-round interview. Submissions finishing 2nd through 10th also receive a $100 participation prize. We will make our hire from the interview group.

## The Project

Build a mobile-first repair cost estimator Progressive Web App (PWA) for Spark Homes' acquisition team. The app is used on site during property walkthroughs to estimate repair costs before submitting an offer.

Acquisition agents walk through a property on a mobile device, check off needed repairs, enter quantities, add photos, and the app calculates a running total. The estimate is exported as a ZIP file containing an Excel cost breakdown and any photos taken during the walkthrough.

**You are building from scratch. A reference implementation is included in the resource folder so you can see what we built use it only as inspiration. Your version must be your own original work and should look and work better.**

### Technical Requirements

- Single self-contained HTML file no build tools, no Node.js server, no frameworks required
- Vanilla JavaScript, HTML, and CSS only (CDN libraries like Tailwind or Chart.js are fine)
- Must work offline  implement a service worker and web app manifest
- Installable as a PWA on Android and iOS (home screen icon, standalone display mode)
- All data persisted in localStorage no backend required, but one is allowed as long as the app works fully offline
- Mobile-first layout, designed for use on a phone or tablet in the field

# Required Features

The app must include all of the following:

### Project Management

- Create, name, and save multiple projects
- Switch between projects without losing data
- Each project stores its own repair selections, quantities, notes, and photos

### Repair Line Items

- 75+ repair line items organized into 5 sections: Interior/General, Kitchen, Bathrooms, Systems & Structure, and Exterior
- Each section contains collapsible groups (19 groups total — see price list for full structure)
- Every group must include a "No Action Needed" option so agents can explicitly mark a group as reviewed with no work required
- Each line item shows: item name, unit type, quantity input, unit cost, and calculated line total
- **Price override:** agents must be able to edit the unit cost for any line item on a per-project basis. There must also be a way to update standard pricing globally so changes roll out across all projects.
- Running cost total visible at all times
- **Add / remove items:** users must be able to add new line items or delete existing ones on a per-item basis.

### Adjustable Room Support

The app must support multiple configurable room types. Agents can add or remove rooms as they walk through the property. This is one of the most important features every house is different, and the room structure must reflect that flexibility.

Each room type comes with its own set of repair groups relevant to that space. Below is the full breakdown of room types and what each covers:

| Room Type | Repair Groups | Notes |
|---|---|---|
| **Bathroom** | Vanity & Countertop, Tub & Shower, Tile | Most flips have 1-3 bathrooms. Each gets its own full set of groups labeled "Bathroom 1", "Bathroom 2", etc. |
| **Kitchen** | Cabinets, Countertops & Tile, Appliances | Usually one per house. Cabinet and appliance condition are major cost drivers on every flip. |

| Room Type | Repair Groups | Notes |
|---|---|---|
| **Interior / General** | Flooring, Paint & Wall Repair, Doors, Pest Control | Applies to the whole house interior. Such as pest control and other house wide expenses |
| **Systems & Structure** | HVAC, Electrical, Structural, Insulation & Drywall | The big-ticket hidden costs. A furnace replacement, full electrical rewire, or foundation repair can each run $3,000-$10,000+. These must never be missed. |
| **Exterior** | Fence, Siding, Windows, Garage, Trees | Curb appeal drives buyer interest and sale price. Exterior condition is evaluated first during any walkthrough. |
| **Bedroom** | Flooring, Paint, Doors, Closet | Added as individual room instances (Bedroom 1, Bedroom 2, etc.). Bedroom-specific repairs are typically cosmetic. |
| **Living / Common Areas** | Flooring, Paint, Doors, Lighting | Living room, dining room, hallways. Often shares repair categories with bedroom but treated as a separate room for accuracy. Currently some of these costs are grouped with interior general but they will need to be decoupled and separated |

Room groups are labeled clearly by instance for example: "Bathroom 1: Vanity & Countertop", "Bathroom 2: Tub & Shower", "Bedroom 1: Flooring". Agents must be able to add and remove individual room instances freely during a walkthrough.

## Progress Tracking

- A progress bar or indicator showing completion percentage
- Progress is counted per group  checking any item in a group marks that group as complete
- Total progress counts all groups across all sections and all room instances

## Photo Capture

- Agents can attach photos to any project directly from the device camera
- Photos display as thumbnails within the project
- Each photo can be removed individually
- Photo capture must work reliably on Android standard file input with capture attribute
- **Serial number parsing:** photos are primarily used to capture serial numbers on HVAC equipment, water heaters, and appliances. If you can extract or attempt to parse a serial number from the photo, that is a significant plus.

## Export

- Export the current project as a ZIP file containing: an Excel spreadsheet with the cost breakdown, and all photos taken during the walkthrough
- The Excel file includes all checked line items, quantities, unit costs, line totals, and a grand total
- The ZIP downloads automatically to the device
- If you come up with a more efficient process, feel free to try to add it.

## Creative Addition (Required)

**Every submission must include at least one feature that was not listed in this brief something you designed yourself.**

It can be anything: a map view, a cost comparison between projects, a contractor contact list, a materials list generator, a deal analyzer that estimates profit margin, voice input, AI-assisted damage detection from photos — whatever you think would make this tool genuinely more useful for a house-flipping team in the field.

*Describe your creative addition in the one-page PDF writeup. This is where strong candidates separate themselves from the pack.*

## The Price List

The attached repair-items.csv contains all 75+ line items with adjusted costs and units. These are the numbers your app should use as default prices. Do not use any other source for pricing, the price list is the single source of truth for this project.

The CSV columns are: id, name, cost, unit. Use the id field to anchor each item in your data model. Users must be able to add new items or delete existing ones on a per-item basis.

## Repair Groups Reference

Your app must include these 19 collapsible groups, organized under the correct section:

| Section | Groups |
|---|---|
| **Interior / General / Common Areas** | Flooring, Paint & Wall Repair, Doors, Pest Control |
| **Kitchen** | Cabinets, Countertops & Tile, Appliances |
| **Bathrooms** | Vanity & Countertop, Tub & Shower, Tile (per bathroom instance) |
| **Systems & Structure** | HVAC, Electrical, Structural, Insulation & Drywall |
| **Exterior** | Fence, Siding, Windows, Garage, Trees |

## What to Submit

Three deliverables, all due by the submission deadline:

1. **Your HTML file:** submit your completed app as a single HTML file (plus any additional static files if needed). Without running a server or installing dependencies
2. **A GitHub repository:** public or private (share access with us). Include a short README explaining your approach, any libraries used, and how to run it locally.
3. **A one-page PDF writeup:** four things: (1) the most interesting design or UX decision you made, (2) what is broken or fragile in your build, (3) your creative addition and why you chose it, and (4) what you would ship next with two more days. Also note what role AI tools played in your development process.

4. Please submit your work via an single email to [James@sparkequitygroup.com](mailto:James@sparkequitygroup.com). For the github repository this should be shared via a link.

## Prizes

| Finish | Prize | What Happens Next |
| --- | --- | --- |
| **1st Place** | $1,000 cash | Final-round interview + top candidate for the role |
| **2nd — 10th Place** | $100 cash each | Final-round interview |
| Unplaced | None | Closing email with feedback |

Total prize pool: $1,900. Prizes are paid via check. We may request a W-9 for tax reporting purposes.

## Evaluation Approach

Three-stage funnel designed to spend the most reviewer time on the highest-signal candidates.

### Stage 1: Triage

The reviewer opens the hosted URL on a phone, spends 5 minutes using the app, and checks whether all required features are present and functional. Submissions missing core functionality (no photo capture, no export, no room support, no progress bar) are cut at this stage.

### Stage 2: Human Review

Two reviewers each score the submission against a rubric across five dimensions:

| Criterion | What We Are Looking For | Weight |
| --- | --- | --- |
| Mobile UX & Polish | Intuitive on a phone, fast, no jank, looks professional | 30% |
| Feature Completeness | All 19 groups, rooms, photos, export, price overrides | 25% |
| Code Quality | Clean, readable, maintainable — not a copy of the reference | 20% |
| PWA & Offline | Installs to home screen, works without network, saves state | 15% |
| Creative Addition | Originality, usefulness, and execution of self-designed feature | 10% |

### Stage 3: Final-Round Interviews

Top 10 submissions each get a 45-minute interview with the Spark Homes team: a live walkthrough of the app followed by a conversation about design decisions, what they would build next, and how they think about shipping real products.

## Timeline

| Date | Milestone |
|---|---|
| 6/23/2026 | Brief released to all 1,600+ applicants |
| 7/14/2026 | Submission deadline  late submissions not reviewed |
| 7/17/2026 | Stage 1 triage complete |
| 7/21/2026 | Top 10 finalists notified; all others receive a closing email |
| 7/21/2026 | Winner announced; $1,000 prize + $100 prizes paid within 5 business days |
| 07/23/2026 | Stage 3 final-round interviews (top 10) |

## Resources

Everything you need is in the shared folder:

| File | Description |
|---|---|
| **Pricing List** | The official price list — 75+ repair items with id, name, cost, and unit. Use these as your app's default prices. |
| **Example App** | Reference implementation — our working version of the app. Look at it for inspiration and scope only. Your submission must be original work. |
| **Spark Group - Logo** | Spark Homes logo for use in your app UI. |
| **Contest Briefing** | This document. |

## Stack and Rules

- Use any CSS framework, UI library, or CDN resource you like
- Use AI tools freely — Cursor, Claude Code, Copilot, whatever you prefer. Note what role AI played in your development process in your writeup
- The final deliverable must be a single HTML file or a small set of static files with no required server-side component
- A backend is optional — if you use one, the app must still work fully offline without it
- The app must work on Chrome for Android and Safari for iOS
- Do not submit the reference implementation as your own — reviewers will know
- Submissions are confidential — do not share this brief or the price list publicly

**We are going to use your app. The submissions that feel fast, reliable, and polished on a phone in a real house are the ones that win. Do not optimize for a screenshot. Optimize for a real walkthrough.**

Questions? Reply to the email this brief was sent from.

Looking forward to seeing what you build.

**Spark Homes**